

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA5607

COURSE OUTCOME COVERED

CO: 5 - Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr. TAMIL SELVI M

QUESTIONS:	Total Marks:	Duration: 90 Minutes
-------------------	---------------------	-----------------------------

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. **Security and Privacy Challenges in Edge Computing for 5G (BL5)**
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

*I **P. Reddy Suhail (1923126711)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Suhail

Survey on Edge Computing Security

Abstract:

With the rise of 5G, autonomous driving, and the Industrial Internet of Things (IIoT), edge computing has gained unprecedented global attention. However, security remains a critical bottleneck restricting its implementation. The new application scenarios, integration of large-scale technologies, and increasing privacy demands present major challenges. This write-up, based on the document [Survey_on_Edge_Computing_Security.pdf](#), explores edge computing security across five core domains: access control, key management, privacy protection, attack mitigation, and anomaly detection, before discussing future directions like edge intelligence and cloud-edge collaboration.

Introduction:

Global data generation is projected to reach 175 ZB by 2025, a surge that exposes the shortcomings of traditional cloud computing—such as high latency, insufficient bandwidth, and severe energy consumption. Edge computing addresses these issues by deploying resources closer to users, significantly reducing data processing delays and minimizing core network loads. Despite these operational benefits, edge nodes face severe vulnerabilities. They are typically deployed in distributed, complex environments with limited computational resources and physical protections, making them prime targets for malicious intrusions.

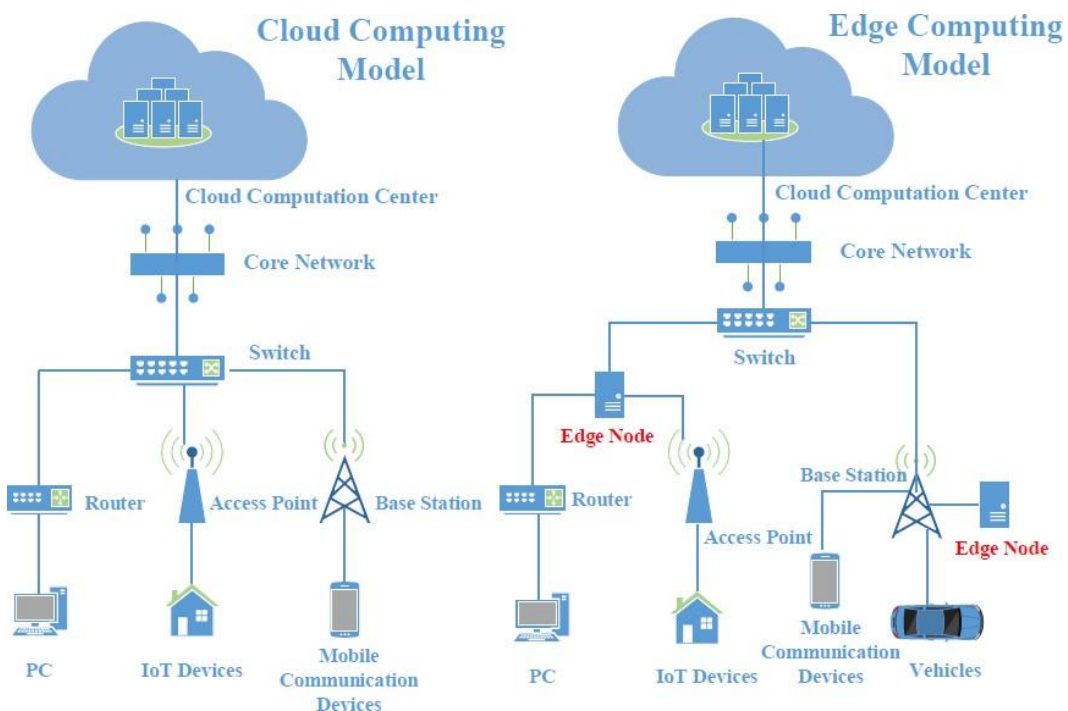


Figure 1: THE COMPARISON BETWEEN EDGE COMPUTING AND CLOUD COMPUTING MODEL

Key Security Challenges:

Research emphasizes five primary security domains within the edge environment:

1. **Access Control:** Edge nodes connect to numerous low-power IoT devices with heterogeneous hardware and protocols. Designing lightweight, service-oriented anonymous identity authentication frameworks is critical.
2. **Key Management:** Traditional PKI and symmetric encryption lack the necessary scalability and efficiency for resource-constrained IoT devices. The industry requires novel, lightweight encryption architectures.
3. **Privacy Protection:** Edge nodes, often considered "semi-honest," handle sensitive user data. Effective mechanisms are required to mask precise user locations, identities, and usage habits from eavesdroppers tracking service migration paths.
4. **Attack Mitigation:** Edge networks expand the attack surface for DDoS attacks. Because edge nodes cannot scale resources dynamically like cloud servers, existing cloud-based mitigation solutions are ineffective.
5. **Anomaly Detection:** The harsh deployment environment makes edge nodes prone to failures. Detecting anomalies and predicting faults in distributed clusters requires advanced, resource-efficient detection mechanisms.

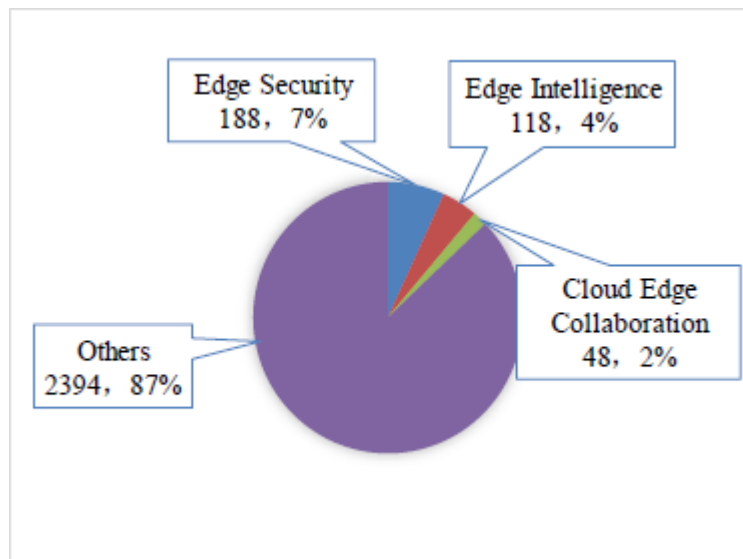


Figure 2: Paper distribution of edge computing during 2016-2020

Conclusion and Future Directions:

The development of edge computing security is currently in its early stages. Moving forward, the industry is trending towards "edge cloud collaboration" and "edge intelligence". Isolated edge computing cannot provide foolproof security due to limited node processing capacities. Future research must focus on the fusion mechanism of edge intelligence and edge security, collaborative models that utilize both cloud power and edge proximity, and the development of edge-embedded lightweight intelligent algorithms.

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Survey on Edge Computing Security

Huang Zeyu

College of Computer Science and Technology
National University of Defense Technology, NUDT
Changsha, China
584411366@qq.com

Xia Geming*

College of Computer Science and Technology
National University of Defense Technology, NUDT
Changsha, China
xiageming@163.com

Wang Zhaohang

College of Computer Science and Technology
National University of Defense Technology, NUDT
Changsha, China
1219383098@qq.com

Yuan Sen

College of Computer Science and Technology
National University of Defense Technology, NUDT
Changsha, China
2604929112@qq.com

Abstract—With the advent of the Internet of Everything, emerging application scenarios including autonomous driving, immersive games based on VR or AR technology, and Industrial Internet of Things have promoted the early development of edge computing technologies. At present, the global mobile communications industry is moving towards 5G. Since 5G is one of the key access technologies to promote large-scale deployment of edge computing, it has made edge computing gain unprecedented attention in the world. However, since the birth of edge computing, edge computing security has been a key issue restricting its implementation and development. The new features of edge computing, the integration of a large number of new technologies, the new application scenarios brought by edge computing, and people's increasing demands for privacy protection have brought great challenges to edge computing security. This paper reviews the current research status of research on edge computing security. By analyzing the security challenges of edge computing in the context of new models, new application scenarios and new technology environments, this paper points out the security problems of edge computing in five aspects, including access control, key management, privacy protection, attack mitigation and anomaly detection. The paper separately discusses the research achievement of the academic community in these five areas and analyzes their advantages and disadvantages. Finally, combining edge cloud collaboration and edge intelligence, the future development direction of edge computing security is discussed and prospected.

Keywords—edge computing; edge computing security; edge cloud collaboration; 5G; edge intelligence

I. INTRODUCTION

Since we entered the era of Internet of everything, the number of various types of networking devices and the amount of data they generate have increased exponentially. According

to IDC's forecast [1], the total amount of global data will increase from 33 zettabytes (ZB) in 2018 to 175 ZB by 2025. More than 49% of the data will be stored in the public cloud environment. According to the data of iResearch in the "Chinese Intelligent Internet of Things (AIoT) White Paper" [2], it is estimated that the number of IoT connections in China will reach nearly 20 billion in 2025, and the number of V2X connected vehicles in China will reach 14% of the total. It can be predicted that if the existing network structure is kept unchanged, the overall communication environment of the Internet will become worse in the future. At that time, the shortcomings of cloud computing, such as high latency, insufficient bandwidth, high energy consumption, data security and privacy risks will be exposed more fully. Therefore, the concept of edge computing has been proposed and has got attention from all walks of life.

Because edge computing deploys computing and storage resources closer to users, it greatly reduces the delay of data processing. On the one hand, edge nodes can directly deal with some tasks that do not need the capacity of cloud server. On the other hand, they can preprocess the tasks and data that need to be transmitted to the cloud server to reduce its bandwidth pressure. At the same time, edge computing can achieve the security and controllability of sensitive data and user privacy by reducing the chance for user data to be transmitted on the core network, deploying encryption and anonymization mechanisms on the edge. Because of these advantages, edge computing has developed rapidly in recent years. Although enterprises have complete solutions in the application scenarios of edge computing such as intelligent security, industrial Internet of things, and intelligent connected vehicles [3], there are still some key problems that hinder the rapid implementation of edge computing and edge computing security is one of them.

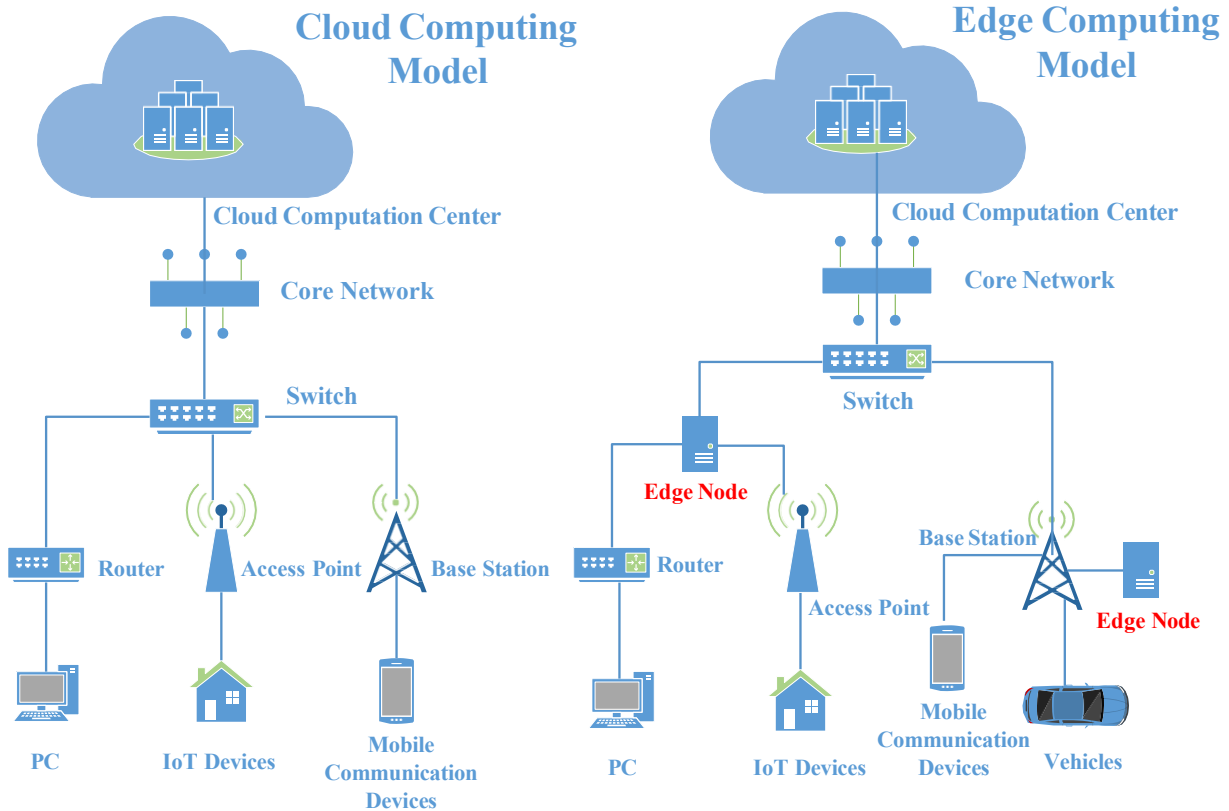


Figure.1. THE COMPARISON BETWEEN EDGE COMPUTING AND CLOUD COMPUTING MODEL

The comparison between edge computing and cloud computing model is shown in Fig. 1. In the traditional cloud computing model, the focus of privacy protection is to ensure that the data is not leaked, so managers can improve the defense ability of cloud server to resist various attacks [4]. While edge nodes are deployed in a distributed manner and are usually placed near the gathering places of people. They have the characteristics of limited resources, complex environment and heterogeneous network, many traditional security mechanisms are difficult to be fully applied on them. Therefore, edge nodes are easy to be invaded by attackers. Because edge nodes will directly connect various IoT devices and wearable devices, they will have stronger environmental awareness than cloud servers, and can get more sensitive information related to users. Because of the vulnerability and complexity of edge node itself, as well as the sensitivity of the data it can access to, the security of edge computing is crucial and challenging.

5G is the key access technology to promote the development of most edge computing applications. As China continues to accelerate the development of the Industrial Internet [5], edge computing will be gradually deployed as a new type of infrastructure with the popularization of 5G. If there is still no complete solution to ensure the security of edge computing, the network Penetration and network attacks similar to Stuxnet [6] may have a direct impact on national security.

In order to solve the existing problems of edge computing security, this paper reviews a large number of published papers on edge computing security from 2016 to early 2020. It summarizes the current research status in the field of edge computing security. Hoping to provide some reference and help for the current and future research in related fields.

II. SECURITY RISKS AND CHALLENGES OF EDGE COMPUTING

In order to solve the new security risks brought by edge computing, some research have been carried out in academia. However, according to the search results in the EI-Compendex database, it can be seen from Fig. 2 and Fig. 3 that since the concept of edge computing was formally proposed in 2016, the number of papers on edge computing security has increased but still not much. In recent years, with the rise of edge intelligence and edge cloud collaboration research, edge computing security research is even more lacking, which reflects the importance and urgency of this research. At present, the problems concerned by academia in the field of edge computing security can be grouped into five categories: access control, key management, privacy protection, attack mitigation, and anomaly detection. Table I shows the sources of security risks for edge computing in the above five fields.

A. Access Control Challenges

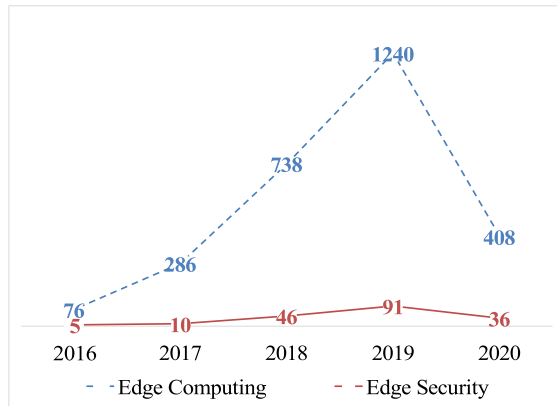


Figure.2. Trend of edge computing papers

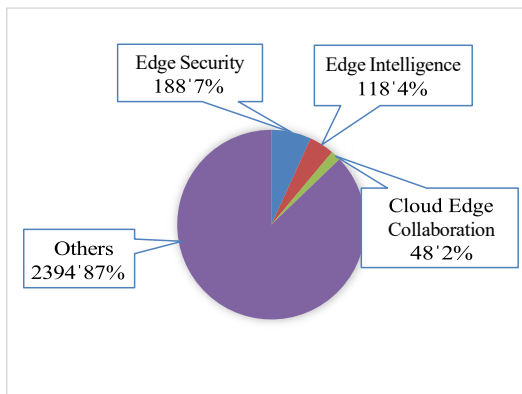


Figure.3. Paper distribution of edge computing during 2016-2020

Edge computing security faces a very complex challenge in access control. Edge nodes will directly connect to a large number of low-power IoT devices. These IoT devices with extremely limited resources, heterogeneous hardware, different communication protocols and difficult to install patches in time are difficult to deploy the security solutions under the traditional cloud computing environment. Therefore, the design of access control mechanism for IoT devices in the edge computing environment will be very critical and challenging.

Identity authentication is also an important issue in the access control of edge computing. In order to avoid revealing the user's true identity information during the identity authentication process, a service-oriented anonymous identity authentication scheme needs to be designed so that the service provider can also confirm the legitimacy of the user without knowing the user's exact identity [7]. At the same time, the integration of edge computing and 5G technology has also increased the complexity of identity authentication.

In addition to the access control challenges faced by edge computing under the current host-centric Internet architecture [8], it may also need to face the challenges of Information-Centric Network (ICN) [8] architecture in the near future. According to the data in the Cisco Visual Network Index [9],

globally, IP video traffic will be 82 percent of all IP traffic (both business and consumer) by 2022, up from 75 percent in 2017. This prompts people to upgrade the existing host-centric network to an information-centric network (ICN) [8], thereby reducing the communication load and delay of the core network. However, due to the existence of intranet caching in ICN networks, traditional streaming media access control schemes are not applicable under this architecture. Successful ICN edge computing deployment relies heavily on an effective decentralized access control mechanism.

B. Key Management Challenges

Although edge computing can enable end-to-end communication of IoT devices using different communication protocols, it still cannot ignore the confidentiality and integrity of data. Safe and controllable key management are the key to ensuring data security and protecting user privacy. The challenge faced by edge computing in key management is: It is necessary to design a secure key management scheme with high scalability, so as to realize the authentication and management of terminal devices with different computing capabilities.

There are two traditional secure communication solutions, one is to distribute unique symmetric encryption keys to each IoT device, and the other is to use the PKI scheme [10]. These two solutions are not applicable to IoT application scenario based on edge computing. The problem of the first solution is that it does not have high scalability. The reason is that in the first solution, each terminal must store the keys of all devices that it needs to communicate with, while IoT devices often have very little storage space. The latter solution is also inappropriate, because the computational capacity of low-power IoT devices is extremely limited, and the use of standard public key infrastructure (PKI) solutions will greatly affect the quality of user experience. Therefore, in order to let edge computing have an applicable key management scheme, one idea is to design a more lightweight and more scalable encryption protocol and encryption primitives based on the current communication security scheme. Another idea is to design a set of communication security schemes with a brand-new architecture. Both research directions are extremely challenging.

C. Privacy Protection Challenges

The biggest challenge for edge computing in terms of privacy protection is that due to the vulnerability of edge nodes, researchers have to assume that they are semi-honest and cannot allow edge nodes to directly obtain the plain text of sensitive information.

In order to prevent criminals from directly obtaining the precise location of the users through edge nodes, it is necessary to design a lightweight positioning encryption mechanism to ensure the security of user's location information. In addition, due to the technical characteristics of edge computing, eavesdroppers can locate users by tracking the migration path of the service [11]. Therefore, it is necessary to design appropriate mechanisms to prevent attackers from acquiring the user's location indirectly.

In addition to the user's location information, user's identity information and interest information also need urgent protection. The edge nodes must not only ensure that users' true identity information is not leaked during the connection and the process of identity authentication [7], but also need to ensure that the type of network slice serving the terminal and the specific type of data transmitted are not leaked. In this way, the user's interest information can be protected. In addition, how to provide a privacy-preserving and efficient data mining scheme on edge nodes is also worth studying. Complex deployment environments, large amounts of sensitive data, and very limited computing resources make edge computing face huge challenges in privacy protection.

D. Attack Mitigation Challenge

At present, an application-level DDoS attack is emerging. It uses IoT devices as attack sources and is more destructive. Two typical case are the Mirai botnet attacks on KrebsOnSecurity [12] and Dyn [13] [14]. With the rapid increase in the number of networking devices in the future, the threat of IoT DDoS attacks will increase. Industrial IoT has greatly expanded the overall attack surface of the system due to the increased connectivity and openness of industrial facilities. The attack threats are extending from the digital to the physical world [15]. However, because a large number of electronic devices in the field control areas have only very limited computation power and lack effective security protection, the current Industrial IoT is very vulnerable to DDoS attacks.

Although the edge nodes can isolate most of the IoT data at the edge of the network and have the ability to detect and intercept attacks in the closest place to the source at the first time, they are facing great challenges in practice. The reason is that the edge node can neither capture the aggregate network traffic required for IoT-DDoS detection, nor can it scale resources needed for mitigation like the elastic cloud [16].

Therefore, it is far from the goal to deploy the existing cloud-based mitigation solutions directly on the edge nodes. It is necessary to redesign a DDoS mitigation solution based on edge computing to overcome the challenges including low detection efficiency, high false alarm rate, long time delay and high computational overhead.

E. Anomaly Detection Challenge

In order to ensure that the applications deployed in the edge nodes can continue to provide high-quality services, anomaly detection must be performed to detect the running state of the edge computing system. Because the deployment environment of edge nodes is relatively harsh, and their own security protection is relatively insufficient, the edge nodes are prone to anomalies. If an anomaly is not handled properly, its impact will propagate from one edge node to other edge nodes, thereby reducing the performance of the entire edge computing system. In addition, once the impact of the anomaly spreads, it is difficult to find the root cause of its occurrence, resulting in additional repair costs and delays in recovery.

However, designing an anomaly detection mechanism for edge computing is very challenging. Edge nodes are deployed in a distributed manner. The size and power consumption of a single node are extremely limited, and its computing and storage resources are far inferior to the cloud computing center. Although the current cloud computing center's anomaly detection system and fault tolerance mechanism are very mature, they will consume a lot of resources, so they cannot be directly moved to the edge computing system. In addition to the challenges of hardware resources, considering the dynamics and heterogeneity of the edge computing system environment, it is also a challenge to connect the abnormal detection results with the hidden failure of the underlying hardware facilities [17]. Detecting and predicting anomalies in a distributed cluster environment is also a current research gap [18]. In summary, designing a reliable anomaly detection mechanism for edge computing still have a long way to go.

TABLE I. THE SOURCES OF SECURITY RISKS FOR EDGE COMPUTING IN THE FIVE FIELDS

Fields	Sources of Risks	Extremely limited terminal resources	Terminals with different communication protocols	New network architecture	Limited edge node resources	Insufficient physical protection of edge nodes	Strong environmental perception of edge nodes
Access Control		√	√	√		√	√
Key Management		√	√			√	
Privacy Protection		√			√	√	
Attack Mitigation		√		√	√	√	√
Anomaly Detection					√	√	

III. ANALYSIS OF THE RESEARCH STATUS OF EDGE COMPUTING SECURITY

In recent years, academic research on edge computing security can be grouped into five categories, including: access control, key management, privacy protection, attack mitigation, and anomaly detection. Although cloud computing already has relatively mature solutions in these fields, many of them are

not suitable for edge computing due to the specificity of edge nodes, such as distributed deployment, limited hardware resources, complex network environment, etc. This prompts researchers to propose more innovative solutions for the characteristics of edge computing. The research hotspots of edge computing security in the above five fields are shown in Table II.

A. Research Status of Access Control in Edge Computing

At present, the research results of edge computing access control are mainly divided into two categories, which are the schemes under the ICN network architecture and the schemes under the non-ICN network architecture.

1) Schemes under the ICN network architecture

In general, there are two main research directions of access control under the ICN network architecture. One is the access control scheme based on user or terminal devices identity authentication, including papers such as [19] [20] [21], the other is the access control scheme based on transmission content encryption, including papers such as [22]. The common disadvantage of early access control schemes based on identity authentication is that the users will experience multiple identity authentication when receiving a complete content, because a series of requests from them will be satisfied by different routers.

In order to solve this problem and the high cost of traditional identity authentication schemes, Prosanta Gope [20] designed a new lightweight privacy-preserving security architecture -- LAAP. The author designs three anonymous authentication protocols using lightweight encryption primitives, so that edge devices can help each other to achieve identity authentication without a central authentication server. However, it is still worth studying whether the protocol can stand the test when malicious users collude with each other. TACTIC, proposed by Reza Tourani et al. [21], is relatively more secure. It delegates authentication and authorization to semi-trusted routers, eliminating the need for always on-line authentication servers. It translates label authentication in subsequent requests of legal users into query operations by a ingenious design, effectively reducing the scale of label authentication and computation overhead effectively. SEAF framework is also an outstanding achievement proposed by Kaiping Xue et al. [19]. In order to reduce the computational cost, SEAF combines the hash chain and group signature technology by taking advantage of the continuity of user requests. So that only the first request of a series of requests needs to be signed for verification with high cost, and the remaining requests use lightweight hash operation for authentication, which only brings slight delay while realizing anonymous authentication DZ

However, the above schemes still have limitations. Firstly, their application scenarios only focus on streaming media services, which is not universal. Secondly, if a user or service provider who has already passed the identity authentication suddenly starts to make malicious behavior that endangers the security of the system for some reason, none of the above schemes can make response in time. At this time, a set of dynamic trust mechanism is needed to supplement the common identity authentication scheme.

Jie Yuan et al. [23] of Beijing University of Posts and Telecommunications proposed a trust framework based on multi-source feedback. They design a lightweight trust evaluation mechanism to promote the cooperation between edge network devices to reduce computing cost and resist malicious feedback attacks. However, according to Zhipeng Gao et al. [24], this trust mechanism ignores the ability of

cloud servers and cannot show the advantage of cloud edge collaboration. Therefore, the team of Zhipeng Gao [24] proposed a lightweight trust mechanism based on cloud edge collaboration, which extracts the trust feature by analyzing the meta interaction data of terminal devices, and enables the edge nodes to make trust evaluation according to the trust feature. The authors improve the LightGBM algorithm. They use convolutional neural network to learn a variety of evaluation algorithms and check the evaluation results of edge nodes on cloud servers. Then the cloud servers can adaptively adjust the trust evaluation strategy of edge nodes according to the checking results. The experimental results show that the accuracy and speed of the trust mechanism are better than other existing methods.

Academic research on access control scheme based on transmission content encryption is not much. Satyajayant Misra et al. [22] proposed AccConf. It is an access control framework based on content encryption. Although it can ensure that only legitimate users can decrypt the corresponding content, and it does not need a central identity authentication server that is always online. It can even solve the problem of privilege revocation quite well. However, this scheme still can't avoid the shortcomings of this kind of research: unauthorized users can still cache the content they can't decrypt. It wastes limited network resources greatly.

2) Schemes under the non-ICN network architecture

There are still many researchers working to solve the access control problem of edge computing under the current Internet architecture. Ranadheer errabelly et al. [25] designed EdgeSec, which is an IoT security service that can be deployed on edge nodes. It can connect to IoT devices with different communication protocols and customize security schemes for them. EdgeSec analyzes the requests sent by each IoT device and simulates sensitive operations in advance to ensure absolute security. However, the service has a high demand for hardware resources.

Jianbing Ni et al. [9] explored the compatibility of edge node authentication and 5G communication new technology. They proposed a service-oriented authentication framework--ES³A. The framework works in IoT edge computing system driven by 5G and supports network slicing technology. In order to protect the user's interest information, the authors design a privacy-preserving slice selection mechanism. It allows the user to access the IoT services anonymously through the agent of the edge node. In addition, session secret keys are generated through negotiation among users, edge nodes and IoT cloud servers.

For a more special disconnected edge environment, Dan Klinedinst et al. [26] proposed a solution to establish a trusted identity. This scheme is oriented to the tactical environment such as first aid, search and rescue, military operations and so on. It does not rely on the Internet and third-party authentication, does not need specific hardware security requirements, and does not need preset certificates. Instead, it realizes identity authentication based on secure key generation and exchange in the field.

At the same time, the popularity of blockchain technology makes researchers try to apply it to edge computing. Shaoyong

Guo et al. [27] proposed a blockchain network built on edge nodes to provide access control for vehicular networks. This scheme uses the blockchain network and digital signature algorithm to authenticate the vehicle's identity. They divide the blockchain network into a three-tier structure to improve the speed of identity authentication. However, if blockchain technology is to be more widely used in edge computing, it still needs to solve the shortcomings of complex network structure, high computing cost, slow authentication speed and so on.

B. Research Status of Key Management in Edge Computing

At present, there are two main research directions of key management scheme for edge computing application environment: one is to carry out lightweight transformation and scalability optimization on the basis of traditional communication security schemes; the other is to design a key management scheme with a new architecture, which has brand new lightweight encryption protocols and encryption primitives.

1) Optimization of traditional communication security solutions

Zhi Li et al. [28] designed a set of key management scheme for fog computing based on hypergraph. The scheme divides the three-layer network structure of edge computing into two subnetworks: cloud-fog subnetwork and fog-user subnetwork. The key management of the two subnetworks is carried out separately to reduce the resource cost and increase the security and scalability of the network. However, in this scheme, all users in the same fog-user subnetwork share the same session key, and the whole network also shares the same session key for cross group data transmission. It brings high security risks.

Sufyan Almajali et al. [29] just put forward an edge computing data transmission encryption scheme, which can solve the security risks exposed in the previous work. This scheme is aimed at the application scenario that the IoT devices transmit data to the cloud servers through the edge nodes. The symmetric key encryption and the PKI encryption system are used to make the edge nodes unable to know the plain text of the conversation between the IoT devices and the cloud servers. At the same time, the anonymity of the IoT devices to the edge nodes is guaranteed. However, whether low-power IoT devices can bear the computational overhead of this scheme still needs to be confirmed by further experiments.

It can be seen that the current lightweight transformation and scalability optimization of traditional communication security schemes are not very satisfactory, and the academic community have to work harder in this direction to get better achievements.

2) Key management scheme of new architecture

It is also a hot research direction to design a new key management scheme with a new architecture for the special application environment of edge computing. Although this direction is helpful for researchers to break out of the constraints of existing communication security schemes, it

also tests their innovation ability for new communication protocols and encryption primitives.

Luca Ferretti et al. [31] expand the current edge computing architecture, so that terminal devices can achieve end-to-end communication only by relying on edge nodes. This architecture assumes that the edge nodes are semi honest, so in order to ensure the communication security, the authors design a novel communication protocol based on symmetric proxy re-encryption algorithm, and finally implement a scalable publish / subscribe communication scheme. This communication scheme only needs to store one encryption key for each terminal device and can perform the communication encryption and decryption operation with very low computing cost. It can ensure the confidentiality of end-to-end communication.

The privacy-preserving SWAN (P²-SWAN) designed by Marc X. Makkes et al. [32] is also an outstanding scheme. It is an edge computing communication security framework driven by homomorphic encryption. The authors optimize the Paillier homomorphic encryption algorithm, so that the communication encryption scheme based on it can run on low-power mobile devices. At the same time, authors also show that the encryption scheme is highly scalable, and only a few edge nodes can bear the data encryption transmission load generated by millions of low-power mobile devices.

Although the design of the edge computing key management scheme with new architecture is very difficult, the experimental results of the current schemes are relatively satisfactory. I believe more surprising achievements will emerge in the near future.

C. Research Status of Privacy Protection in Edge Computing

Although almost all research on edge computing security is related to privacy protection more or less, some privacy protection problems still need to be solved with highly targeted solutions, such as user location privacy protection and identity privacy protection.

In the future, mobile devices can rely on edge computing and 5G technology to achieve high-precision positioning. So in order to reduce the security risks caused by location information leakage, Han Jiang et al. [33] put forward two privacy protection positioning protocols for edge computing based on triangle positioning algorithm, multi-point positioning algorithm and Paillier homomorphic encryption. The two protocols protect the location data directly. The security of the protocol depends on the semantic security of Paillier homomorphic encryption.

At the same time, in order to avoid eavesdroppers indirectly capturing the user's location by tracking the migration path of services between edge nodes, Ting He et al. [11] designed a chaff service to fight against these eavesdroppers. They studied a variety of service control strategies to confuse and interfere with eavesdroppers' tracking. The authors also propose extended strategies for the advanced eavesdroppers who can detect the control strategy of the chaff. In the extended strategies, the randomization method is used to add disturbance to the control strategies proposed

earlier. The efficacy of the extended strategies is confirmed through both synthetic and trace-driven simulations.

In order to prevent users from disclosing their interest information while enjoying VOD service, Simon Da Silva et al. [34] proposed PrivaTube. It is a scalable and cost-effective VOD solution. The scheme makes full use of the trusted execution environment on the edge node and the client. It uses forged requests to confuse the users' real access habits, so as to reduce the risk of personal interest information disclosure. Siddhartha Kumar et al. [35] proposed a private information retrieval scheme based on generalized Reed-Solomon code to prevent users from disclosing their interest information during private information retrieval on edge nodes. This scheme can protect users' interest information even when multiple small cell base stations are colluding with each other.

At present, although the research on privacy protection of edge nodes is hot, many achievements ignore the load problem brought by privacy protection scheme to edge nodes. If edge nodes spend too much resource to protect the privacy of users, it will inevitably affect their quality of service, so the lightweight privacy protection schemes will be a great research direction in the future.

D. Research Status of Edge Computing Attack Mitigation

Although edge computing may encounter a variety of network attacks, most of the current research achievements are aimed at IoT DDoS attacks. In order to make full use of edge nodes to deal with the threat of IoT DDoS attacks, Ketan Bhardwaj et al. [36] proposed a DDoS mitigation scheme. The scheme aggregates the traffic distribution of multiple edge nodes in three steps to detect DDoS attacks. Although the scheme can detect DDoS attacks faster in theory, if the edge node captures too little traffic, it will be difficult to find DDoS attacks with very scattered attack sources. Qiao Yan et al. [37] proposed a multi-layer DDoS mitigation framework, including edge computing layer, fog computing layer and cloud computing layer. The first two layers are responsible for the capture of network traffic information. The cloud servers are responsible for the aggregation of network traffic information and DDoS attack detection. The detection results will be fed back to the fog computing layer. Fog servers will be responsible for dealing with attack traffic. The framework uses Snort rules to detect DDoS attacks and uses switches driven by SDN to mitigate attacks.

Based on the above research achievements, Zhou Lying et al. [15] designed a set of IoT DDoS mitigation scheme for the industrial IoT environment. It still uses three-tier architecture to carry out the work. The firewall on the edge node will reversely filter the attack traffic packets based on the known attack traffic signature; the fog node will initially check the limited network traffic captured by itself according to the

rules, mark the new and obvious attack traffic, then send its signature to the edge nodes for filtering; the cloud servers are responsible for collecting the information from multiple sources and analyzing it much more meticulously to make a more accurate judgment. Experimental results show that compared with the previous studies, this scheme can not only minimize the false alarm rate to the greatest extent, but also get a faster response speed. Vafa Andalibi et al. [38] have studied the methods of resolving various network attacks from the IoT devices themselves. They try to set a rate limit for each IoT device in the manufacturer usage description (MUD). It can prevent the IoT devices from participating in DDoS attacks from the source. However, whether this scheme is feasible in reality remains to be further verified by experiments.

From the current research results, most of the IoT DDoS mitigation schemes have begun to use the idea of cloud edge collaboration to improve the response speed and identification accuracy of the system. This defense idea can comprehensively utilize the capabilities of each layer of the network to deal with various network emergencies in a multi-layer collaborative way. It can be predicted that the idea of cloud edge collaboration will be further developed in the field of edge computing attack mitigation and play an important role in coping with other forms of network attacks.

E. Research Status of Edge Computing Anomaly Detection

In order to discover and troubleshoot the edge node anomalies in the complex heterogeneous edge environment in time, the academic community has carried out some research and get many achievements. Areeg Samir et al. [39] proposed an anomaly detection and prediction model based on Hidden Markov model for the edge environment to solve the mapping problem between the detection results and the underlying infrastructure failure. The model comprehensively analyzes the anomaly of the edge node by monitoring the resource utilization rate, data throughput and the response time of the container on edge nodes. At the same time, the model can also carry out anomaly prediction to optimize system availability and performance. Mbarka Soualhia et al. [40] proposed a fault detection and prediction framework for edge infrastructure based on supervised machine learning and statistical technology. In addition to the anomaly detection of edge nodes themselves by monitoring their own key operation data, the framework also enables the cluster master node to detect and predict the fault of the first-line high load slave nodes. The master node can schedule the workload ahead of time for the execution of the recovery mechanism. Nour Moustafa et al. [41] proposed an anomaly detection mechanism based on outlier Dirichlet. In view of the current anomaly detection based on machine learning mechanism is vulnerable to attacks from APT

TABLE II. RESEARCH HOTSPOTS IN EACH LEVEL OF EDGE COMPUTING SECURITY

Research Fields	Research Hotspots	
Access Control	Schemes under the ICN network architecture	● Schemes based on terminal equipment identity authentication

	Schemes under the non-ICN network architecture	• Schemes based on transmission content encryption • Access control schemes based on trust mechanism • IoT security framework based on edge computing
Key Management	• Optimization of traditional communication security solutions • Key management schemes with new architecture	
Privacy Protection	• User's location privacy protection schemes • Privacy protection of user identity and interest information • Security Solution for Edge Data Mining	
Attack Mitigation	• IoT DDoS attack mitigation solutions • Universal attack mitigation schemes	
Anomaly Detection	• Anomaly Detection and Prediction Schemes for Edge Cloud Collaboration • Anti-attack research for anomaly detection	

attackers confrontational, it uses an adversarial statistical learning mechanism to strengthen the anomaly detection model's resistance against data poisoning attacks. Ensure that it can detect anomalies in large-scale dynamic network normally in the face of attacks targeting the process of machine learning.

From the current research achievements, edge computing anomaly detection also presents the trend of edge cloud collaboration. When edge nodes perform lightweight anomaly detection, cloud servers can implement machine learning algorithm and analyze the state of edge node more deeply. It can not only supervise the edge nodes' anomaly detection, but also realize the anomaly prediction for edge nodes. In addition, the anti-attack research on this type of anomaly detection mechanism is also gradually deepened. I believe that the research achievements of edge anomaly detection based on edge intelligence will be more and more in the near future.

IV. FUTURE DEVELOPMENT DIRECTION OF EDGE COMPUTING SECURITY

From the above research status of edge computing security, more and more research results are beginning to show the trend of edge cloud collaboration. Combining with the overall evolution of the edge computing industry, all aspects of edge computing security research will develop towards edge cloud collaboration and edge intelligence in the future. On the one hand, isolated edge computing models cannot provide truly effective and safe services. The relatively limited processing capacity of edge nodes, the highly complex edge network environment, and the extremely high mobility of terminal devices in the edge environment restrict edge computing, which cannot provide complete security services by their own capabilities. The bottleneck encountered by edge computing security in this aspect can be clearly seen from the research status. It will become a development trend to combine cloud computing and edge computing for edge cloud collaboration, comprehensively utilize the capabilities of each layer of the network to improve edge performance and then provide more efficient, reliable, and flexible security services. On the other hand, the combination of edge computing and artificial intelligence, edge intelligence [42] has become a new frontier hotspot. Through the mutual empowerment of edge computing and artificial intelligence, it can meet the new application needs of real application scenarios [43]. Edge computing can provide infrastructures and real-time data for artificial

intelligence sinking to the edge of the network [44], making intelligence develop from the cloud to the users. In this way, the "last mile" problem of artificial intelligence landing can be solved. At the same time, implementing artificial intelligence algorithms at the edge can solve many key issues in edge computing, and security is one of them.

At present, some research results have clearly applied edge intelligence to edge computing security. Zengguang Liu et al. [45] designed a novel intrusion detection and defense method for CPSS (Cyber-Physical-Social Systems) low-rate DDoS (LR-DDoS) scenarios in the edge computing environment. This method can extract local sensitive features and use deep convolution neural network (DCNN) to automatically learn the best feature distribution of the original data. Then, use a deep Q network (DQN) to make decisions on attack prevention. Experimental results show that this method has higher detection accuracy and faster response speed than SVM and K-means. It also has a certain detection rate for unknown new network attacks. In terms of privacy protection, Zhou Zhi et al. [46] proposed a data privacy protection scheme that combines edge intelligence with federated learning. Edge intelligence can also be used for edge security service placement and situation awareness.

However, edge computing security and edge intelligence need to be mutually empowered. Although edge intelligence can solve many problems encountered in current edge computing security, the realization of edge intelligence also depends on the proper solution of the fundamental problems of edge security. First of all, edge intelligence will use a lot of private data during model training. It faces a greater risk of privacy leakage [47]. Secondly, as an important digital asset, the edge intelligent model is easy to be stolen and cracked by malicious users after its deployment to the edge. It may result in serious economic losses. In the edge cloud collaboration mode, the realization of edge intelligence not only depends on the real-time processing of terminal perception data and user uploaded data by the edge nodes, but also requires the cloud computing center to perform more detailed analysis and make decisions on the pre-processed data sets uploaded by the edge nodes. This process involves the transmission of a large amount of sensitive data, which brings great challenges to the security of edge intelligence in all aspects, including equipment, network, data and applications[48].

To sum up, in view of the current research status of edge computing security and the development trend of edge

computing to edge intelligence, we propose three future research hotspots for the research on the security mechanisms and key technologies of edge intelligence and edge cloud collaboration:

- The fusion mechanism of edge intelligence and edge security. Edge intelligence need to be guaranteed by edge security and edge security can be improved by edge intelligence. The fusion mechanism between edge intelligence and edge security is a key scientific issue. The mutual empowerment of edge computing and artificial intelligence is worthy of further study. Their fusion level and coupling position in the edge computing system, the mutual embedded interface and API in the edge network, etc., involve many basic theories and key technologies, such as network system, network protocols, security protocols and data analysis, etc., which need to be further explored by researchers.
- The collaboration model in the edge-cloud collaboration architecture. Edge security and edge intelligence can be enhanced by edge cloud collaboration. The advantages of edge computing compared with cloud computing can also be retained or even strengthened by edge cloud collaboration. The collaboration model is a key scientific issue in edge cloud collaboration architecture. How to realize the hierarchical edge cloud collaboration model and the distributed edge-edge collaboration model, what is their combination and switch mechanism under a unified architecture, and what is their implementation mechanism on the edge of the network all need to be carefully studied by researchers.
- Edge embedded lightweight intelligent algorithm and multi-party security protocols. In the complex environment of edge computing, the edge security architecture is based on the limited resources of edge nodes and edge networks, in which the lightweight intelligent algorithm and multi-party security protocols implemented by the edge are the key scientific issues. Researchers need to design and implement distributed intelligent algorithms that can be effectively executed by edge nodes, distributed and lightweight multi-party security protocols that can be effectively run by edge networks, and the embedded mechanism of the two. This research involves lightweight artificial intelligence algorithms, group intelligence algorithms, secure multi-party computing based on blockchain, security protocols, cryptography and other aspects of the basic theories and key technologies. It is a complex scientific issue that need to be carefully studied.

V. CONCLUSION

Although the research on edge computing security is increasing gradually, the overall development is still in the early stage. In the current development trend of edge computing to edge intelligence, edge computing security lacks representative research achievements in trust evaluation algorithms and authentication mechanisms, key management

mechanisms and lightweight security protocols, identity privacy and behavior privacy protection technology, multi-layer collaborative intelligent anomaly detection algorithms, edge cloud collaborative intelligent attack mitigation mechanisms, etc. After investigating a large number of edge computing security papers, this paper first introduces the background knowledge of edge computing. Secondly, the paper analyzes the security challenges of edge computing from five aspects. Thirdly, the paper classifies the current main research achievements of edge computing security into five categories and respectively introduces the current situation of security research in these five fields in academic. Finally, the possible research directions in the future are prospected. It is expected that edge computing can be implemented faster and people can enjoy the convenience brought by edge intelligence safely by solving these hot issues of edge computing security in the future.

REFERENCES

- [1] David Reinsel, John Gantz, John Rydning. The Digitization of the World-From Edge to Core [EB/OL]. [2018-11]. <https://www.seagate.com/files/www-content/our-story/trends/files/idc-seagate-dataage-whitepaper.pdf>
- [2] iResearch. China Artificial Intelligent Internet of Things (AIoT) White Paper [EB/OL]. [2020-02-27]. http://report.iresearch.cn/report_pdf.aspx?id=3529
- [3] Chinese Association of Automation. Edge Computing 2019 Album-Editorial [J]. Automation Panorama, 2019, 11:3
- [4] Liehuang Z , Feng G , Meng S , Yandong L , Baokun Z , Hongliang M , et al. Survey on Privacy Preserving Techniques for Blockchain Technology[J]. Journal of Computer Research and Development, 2017.
- [5] H Yu, P Zeng, Z Weng, H Huang, C Chen, Y Tao, et al. Edge Computing Security White Paper [EB/OL]. Edge Computing Consortium and Alliance of Industrial Internet. [2019-11]. <http://www.cbdi.com/image/site2/20191128/f42853157e261f49c5df5b.pdf>
- [6] General Office of the Ministry of Industry and Information Technology of the People's Republic of China. Notice of the General Office of the Ministry of Industry and Information Technology on Promoting the Development of Industrial Internet [EB/OL]. The website of Ministry of Industry and Information Technology of the People's Republic of China. [2020-03-20]. http://www.cac.gov.cn/2020-03/20/c_1586243822373053.htm
- [7] NIST. Report on lightweight cryptography [EB/OL]. [2016-09-16]. <http://csrc.nist.gov/publications/drafts/nistir-8114/nistir8114draft.pdf>
- [8] Kamouskos S . Stuxnet Worm Impact on Industrial Cyber-Physical System Security[J]. 2011.
- [9] Ni J , Lin X , Shen X . Efficient and Secure Service-oriented Authentication Supporting Network Slicing for 5G-enabled IoT[J]. IEEE Journal on Selected Areas in Communications, 2018:1-1.
- [10] Roman R , Zhou J , Lopez J. On the features and challenges of security and privacy in distributed internet of things[J]. Computer Networks, 2013, 57(10):2266-2279.
- [11] He T, Ciftcioglu E N, Wang S, Chan K S. Location Privacy in Mobile Edge Clouds: A Chaff-Based Approach[J]. IEEE Journal on Selected Areas in Communications, 2017, 35(11): 2625-2636.
- [12] KrebsOnSecurity hit with record DDoS [EB/OL]. [2020-03-20]. <https://krebsonsecurity.com/2016/09/krebsonsecurity-hit-with-record-ddos/>
- [13] Dyn analysis summary of Friday October 21 attack [EB/OL]. [2020-03-20]. <http://dyn.com/blog/dyn-analysis-summary-of-friday-october-21-attack/>
- [14] Breaking down mirai: An iot DDoS botnet analysis[EB/OL]. [2020-03-20]. <https://www.incapsula.com/blog/malware-analysis-mirai-ddos-botnet.html>

- [15] Zhou L, Guo H, Deng G. A Fog Computing Based Approach to DDoS Mitigation in IIoT Systems[J]. *Computers & Security*, 2019: 51-62.
- [16] Ketan Bhardwaj, Joaquin Chung Miranda, Ada Gavrilovska. Towards Iot-ddos Prevention Using Edge Computing[C]. *Usenix Workshop on Hot Topics in Edge Computing (HotEdge 18)*, Boston MA, 2018.
- [17] A. Samir and C. Pahl. A Controller Architecture for Anomaly Detection, Root Cause Analysis and Self-Adaptation for Cluster Architectures[C]. *Intl Conf Adaptive and Self-Adaptive Systems and Applications*, 2019.
- [18] Du Q, Xie T, He Y. Anomaly Detection and Diagnosis for Container-Based Microservices with Performance Monitoring[C]. *International Conference on Algorithms and Architectures for Parallel Processing*. Springer, Cham, 2018.
- [19] K. Xue, He P, Zhang X, Xia Q, Yue H, Wu F, et al. A Secure, Efficient, and Accountable Edge-Based Access Control Framework for Information Centric Networks[J]. *IEEE/ACM Transactions on Networking*, vol. 27, no. 3, pp. 1220-1233, June 2019.
- [20] Gope, P. LAAP: Lightweight anonymous authentication protocol for D2D-Aided fog computing paradigm[J]. *Computers & Security*, 86. pp. 223-237.
- [21] Reza Tourani, Ray Stubbs, Satyajayant Misra. TACTIC: Tag-based Access Control Framework for the Information-Centric Wireless Edge Networks[C]. 2018 IEEE 38th International Conference on Distributed Computing Systems (ICDCS), Vienna, 2018, pp. 456-466.
- [22] Misra S, Tourani R, Natividad F, Mick T, Majd N E, Huang H. AccConF: An Access Control Framework for Leveraging In-Network Cached Data in the ICN-Enabled Wireless Edge[J]. *IEEE transactions on dependable and secure computing*, 2019, 16(1):5-17.
- [23] Yuan J, Li X. A multi-source feedback based trust calculation mechanism for edge computing[C]. *conference on computer communications workshops*, 2018: 819-824.
- [24] Z. Gao, Xia C, Jin Z, Wang Q, Huang J, Yang Y et al. A Lightweight Trust Mechanism for Cloud-Edge Collaboration Framework. 2019 IEEE 27th International Conference on Network Protocols (ICNP), Chicago, IL, USA, 2019, pp. 1-6.
- [25] Sha K, Errabelly R, Wei W, Yang T A, Wang Z. EdgeSec: Design of an Edge Layer Security Service to Enhance IoT Security[C]. *IEEE International Conference on Fog & Edge Computing*. IEEE, 2017.
- [26] Echeverria S, Klindinst D, Williams K, Lewis G A. Establishing Trusted Identities in Disconnected Edge Environments[C]. 2016 IEEE/ACM Symposium on Edge Computing (SEC). ACM, 2016.
- [27] S. Guo, X. Hu, Z. Zhou, X. Wang, F. Qi and L. Gao. Trust access authentication in vehicular network based on blockchain[J]. *China Communications*, vol. 16, no. 6, pp. 18-30, June 2019.
- [28] Li Z, Liu Y, Liu D, Li C, Cui W, Hu G. A Key Management Scheme Based on Hypergraph for Fog Computing[J]. *China Communications*, 2018, 15(11):164-176.
- [29] Almajali S, Salameh H B, Ayyash M, Elgala H. A Framework for Efficient and Secured Mobility of IoT Devices in Mobile Edge Computing[C]. *Third IEEE International Conference on Fog & Mobile Edge Computing*. IEEE, 2018.
- [30] E. Gyamfi, J. A. Ansere and L. Xu. ECC Based Lightweight Cybersecurity Solution For IoT Networks Utilising Multi-Access Mobile Edge Computing. 2019 Fourth International Conference on Fog and Mobile Edge Computing (FMEC), Rome, Italy, 2019, pp. 149-154.
- [31] Ferretti L, Marchetti M, Colajanni M. Fog-based Secure Communications for Low-power IoT Devices[J]. *ACM Transactions on Internet Technology*, 2019, 19(2): 1-21
- [32] Makkes M X, Uta A, Das R B, Bozdog V N, Bal H. P²-SWAN: Real-Time Privacy Preserving Computation for IoT Ecosystems[C]. *IEEE International Conference on Fog & Edge Computing*. IEEE, 2017.
- [33] Jiang H, Wang H, Zheng Z, Xu Qiuliang. Privacy preserved wireless sensor location protocols based on mobile edge computing[J]. *Computers & Security*, 2018.
- [34] Silva S D, Mokhtar S B, Contiu S, Daniel N, Laurent R, Etienne R. PrivaTube: Privacy-Preserving Edge-Assisted Video Streaming[C]. *international middleware conference*, 2019: 189-201.
- [35] Kumar S, Amat A G I, Rosnes E, Senigaglia L. Private Information Retrieval From a Cellular Network With Caching at the Edge[J]. 2018.
- [36] Ketan Bhardwaj, Joaquin Chung Miranda, Ada Gavrilovska. Towards IoT-DDoS prevention using edge computing[C]. *Proceedings of USENIX workshop on hot topics in edge computing, HotEdge 18*, 2018.
- [37] Yan Q, Huang W, Luo X, Gong Q. A Multi-Level DDoS Mitigation Framework for the Industrial Internet of Things[J]. *IEEE Communications Magazine*, 2018, 56(2):30-36.
- [38] Vafa Andalibi, DongInn Kim, L. Jean Camp. Throwing MUD into the FOG: Defending IoT and Fog by expanding MUD to Fog network[J]. *Proceedings of USENIX workshop on hot topics in edge computing, HotEdge 19*, 2019.
- [39] A. Samir and C. Pahl. Detecting and Predicting Anomalies for Edge Cluster Environments using Hidden Markov Models. 2019 Fourth International Conference on Fog and Mobile Edge Computing (FMEC), Rome, Italy, 2019, pp. 21-28.
- [40] Mbarka Soualhia, Chunyan Fu, Foutse Khomh. Infrastructure Fault Detection and Prediction in Edge Cloud Environments[C]. *Proceedings of the 4th ACM/IEEE Symposium on Edge Computing, (SEC 2019)*, Arlington, VA, USA, November 7-9, 2019.
- [41] Moustafa N, Choo K R, Radwan I, Camtepe S. Outlier Dirichlet Mixture Mechanism: Adversarial Statistical Learning for Anomaly Detection in the Fog[J]. *IEEE Transactions on Information Forensics and Security*, 2019, 14(8): 1975-1987.
- [42] S. Deng, H. Zhao, W. Fang, J. Yin, S. Dustdar and A. Y. Zomaya, "Edge Intelligence: The Confluence of Edge Computing and Artificial Intelligence," in *IEEE Internet of Things Journal*. ArXiv, 2019.
- [43] Z Zhou, S Yu, X Chen. Edge intelligence: a new nexus of edge computing and artificial intelligence [J]. *Big Data Research*, 2019, 005(002):P.53-63.
- [44] K. Bhardwaj, N. Suda and R. Marculescu. EdgeAI: A Vision for Deep Learning in IoT Era. in *IEEE Design & Test* 2019.
- [45] Liu Z, Yin X, Hu Y. CPSS LR-DDoS Detection and Defense in Edge Computing Utilizing DCNN Q-Learning[J]. *IEEE Access*, 2020: 42120-42130.
- [46] Zhou Z, Chen X, Li E, Zeng L. Edge Intelligence: Paving the Last Mile of Artificial Intelligence with Edge Computing[J]. *Proceedings of the IEEE*, 2019.
- [47] Y. Lu, X. Huang, Y. Dai, S. Maharjan and Y. Zhang. Differentially Private Asynchronous Federated Learning for Mobile Edge Computing in Urban Informatics. in *IEEE Transactions on Industrial Informatics*, vol. 16, no. 3, pp. 2134-2143, 2020.
- [48] Y. Tao, P. Xu and H. Jin. Secure Data Sharing and Search for Cloud-Edge-Collaborative Storage. in *IEEE Access*, vol. 8, pp. 15963-15972, 2020.